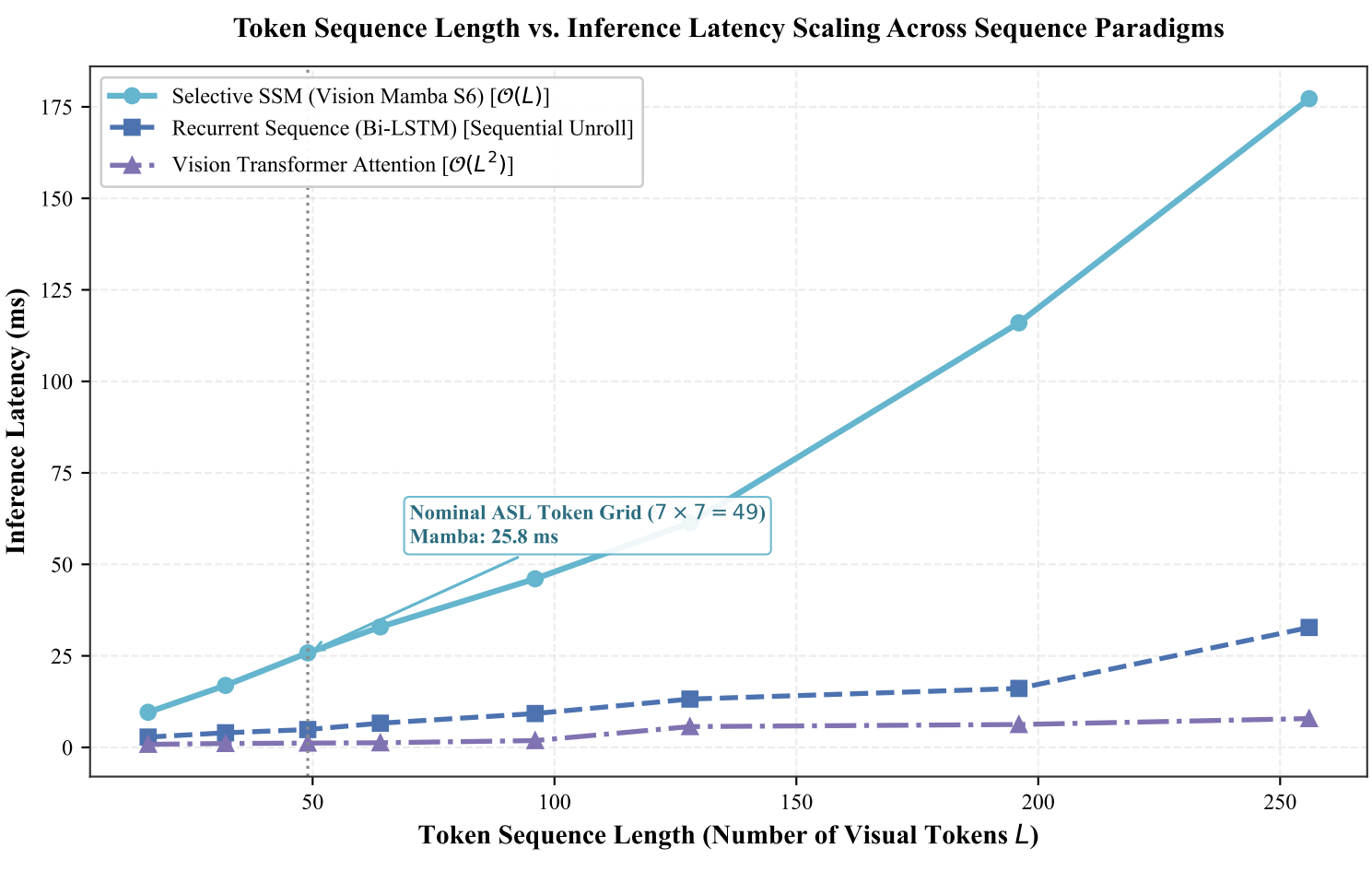
Token Sequence Length vs. Inference Latency Scaling Across Sequence Paradigms
Selective SSM (Vision Mamba S6) [$\mathcal{O}(L)$]
Recurrent Sequence (Bi-LSTM) [Sequential Unroll]
Vision Transformer Attention [$\mathcal{O}(L^2)$]
Nominal ASL Token Grid ($7 \times 7 = 49$)
Mamba: 25.8 ms
Inference Latency (ms)
Token Sequence Length (Number of Visual Tokens L)
0
25
50
75
100
125
150
175
50
100
150
200
250